

OFFLINE SEED FROM A DECK

AVB 52 card generator

A pair of single-file HTML tools. You shuffle a real 52-card deck, type what you deal, and the page turns that deal into a BIP39 recovery phrase. Nothing leaves the machine.

What it is

Open **avb-52-card-generator.html** and choose Basic or Full. Both files use the same card ritual. They differ only in where they stop.

Basic	Cards become 12 or 24 BIP39 words. The file stops there. No addresses, no QR, no keys in memory. For people who write the words on paper or steel and restore elsewhere.
Full	Same ritual, then the first three native SegWit (bc1) receive addresses, the account zpub at m/84'/0'/0', and a SeedQR. For people who want to check addresses before they leave the air-gap box. That screen is the seed.

Eighteen-word phrases are not offered. There is no network, no account, and no browser storage.

What you need

One complete 52-card deck. An offline computer. A current browser. Paper or steel. The HTML file copied onto that computer — not opened from a live website. After the page loads, the self-test banner must be green. If it is red, stop.

How the ritual works

- Length first.** Choose 12 or 24 words before the first card. Changing length later clears the deal. The 12-word phrase is the first half of the matching 24-word phrase. Looking at twelve words and then extending would leak half the final seed.
- How you read the cards.** Ranks only is the default: type A, 2–9, T (or 10), J, Q, K, ignore suits, deal **45** ranks. Full cards is shorter: rank plus suit (AH, 7C, 10D, QS), deal **30** cards. You never mix 45 with 30.
- Shuffle and deal.** Riffle at least seven times or wash face down for thirty seconds. Deal face up. Type what you see. The line under the box shows how the page read your text. **1** becomes Ace. **10** becomes T.

12 words: one shuffle, one deal. Stop when the meter is full.

24 words: two deals of the same count (45+45 or 30+30). Each deal starts from a **full** 52-card deck. When deal one is full the box refuses more cards. Gather every card, including those already face up. Shuffle again. Press Confirm deal one (that box greys out). Deal the same count again. Press Finished round 2. The words appear blurred; unblur them and write them down. Round two is never the leftover pile.

Notation

Card	Type
Ace	A or 1
Two through nine	2 ... 9
Ten	T or 10
Jack Queen King	J Q K
Suits (full-card mode)	C clubs, D diamonds, H hearts, S spades

A repeat inside one deal is rejected. After the reshuffle a card from deal one may appear again.

What the page computes

Each card is a draw without replacement from a known deck. That narrows an interval. When the interval is small enough to lock 128 bits (12 words) or 256 bits (24 words), those bits are the BIP39 entropy. A SHA-256 checksum is appended and the bits are read eleven at a time against the official English wordlist.

The orange bits on the grid are that checksum: calculated from the entropy, not extra randomness. The same hex entropy pasted into another BIP39 tool that accepts raw entropy (Ian Coleman and others) should produce the same words. Always restore on a second device before you send funds.

Full file only

After the phrase exists, the Full file can derive zpub at m/84'/0'/0', receive addresses m/84'/0'/0'/0 through /2, and a Standard SeedQR for an offline signer such as SeedSigner. No BIP39 passphrase. No Web Crypto. Do not photograph the QR. Close the browser when finished.

What it does not do

It does not talk to the network. It does not store the phrase. Basic never derives keys. Neither file is a wallet you spend from. The self-test proves this copy's math matched built-in vectors; it does not prove the file was never edited. Compare the SHA-256 of the file on disk with the list in the README if you need that.

SHA-256 of the shipped files

```
cb22bbdfe78cd2629bc6d8e61db6d9152293189d0b0233c66c6199a694ea2e6  chooser
331a476f1b87aebba982e6f72b74bad7827fa6f2955aef6bec53c7dcd18d242f  basic
37642d61986634c129ccc150c11f552a49d450bc941b557b84768e639ce91522  full
```